

Forest Engineering

Land Sciences

May, 2026



**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Short Title: Forest Engineering
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author TKENT
Credits 5

Level: Intermediate

Description of Module / Aims

In this module students will learn about technical solutions to the most important forest operations, i.e. road construction, drainage, ground preparation and planting. During the module students will be required to assess the environmental consequences and health and safety requirements of different technical solutions. Students will also learn about the increasingly important use of wood as a renewable fuel, and the machinery and methods of converting forest residues and small trees to wood particles.

Programmes

FORS-0016 BSc in Forestry (WD_SFORS_D)

3 / 6 / M

Indicative Content

- Land drainage: types of soil water; principles of drainage; methods of drainage and design of drains; subsoiling, ripping, mole drains, gravel moles, and deep ploughing; environmental concerns in relation to forest drainage
- Forest establishment and maintenance: mechanical site clearing and ground preparation; mechanical planting and vegetation control; pruning and non-commercial thinning
- Forest roads: road and site classification; forest road design; standards for forest roads; road construction; road degradation and maintenance; environmental concerns in relation to forest roads
- Road haulage of forest products: truck configuration; permissible loads; route planning optimisation and agreed routes
- Comminution of wood: characteristics of forest assortments available for comminution; wood chipping, shredding and grinding equipment; wood particle products quality parameters
- Wood energy: characteristics of wood as a fuel; wood burning boiler technologies; quantification and conversion of woodfuel

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Design the drainage requirements, the appropriate method of drainage, and drainage maintenance requirements for a range of land types.
2. Analyse the processes used in forest establishment and maintenance and recommend the appropriate machines and methods for a range of forest sites.
3. Establish the main processes involved in forest road planning, design, construction and maintenance and assess road suitability for different types of timber trucks.
4. Investigate the use of forest resources for wood energy and the machinery and methods used in wood fuel production.
5. Plan forest operations in a manner that complies with best health and safety practice and environmental guidelines.
6. Produce a report and map containing relevant information on site conditions, environmentally sensitive areas and social issues in relation to road building and design a forest road including construction cost.
7. Prepare a forest land development plan, including drainage design, ground preparation and establishment methods and costs.
8. Analyse a wood fuel supply chain and the woodfuel requirements and energy costs of a wood-fired boiler.

Learning and Teaching Methods

- Lectures
- Fieldwork

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3, 4, 5
Continuous Assessment	50%	
<i>Project</i>	15%	6
<i>Project</i>	15%	7
<i>Project</i>	20%	8

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	24	
Independent Learning	75	

Essential Material

- Anon., 1. *_Forestry and Water Quality Guidelines_*. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2000.
- Anon., 2. *_Forestry Schemes Manual_*. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2011.
- Collins, K., G. Gallagher, J. Gardiner, E. Hendrick and D. McAree. *_Code of Best Forest Practice - Ireland_*. Ireland: Forest Service, Department of Agriculture, Food and the Marine, 2000.
- Forest Industry Transport Group. *_Managing Timber Transport Good Practice Guide_* by Lyons, J., F. Russell, M. Joyce and G. Devlin.. Dublin, Ireland. 2014.
- Forest Service, Department of Agriculture, Food and the Marine. *_Forest Roads Scheme 2014-2020_* Ireland. 2015.
- Irish Forest Industry Chain and Forest Industry Transport Group. *_Road Haulage of Round Timber Code of Practice_* by Joyce, M. and E. Daly.. Dublin, Ireland. 2012.
- Kent, T., P. Kofman and E. Coates. *_Harvesting wood for energy - Cost-effective woodfuel supply chains in Irish forestry_*. Dublin, Ireland: COFORD, 2011.
- Mulqueen, J., M. Rodgers, E. Hendrick, M. Keane and R. McCarthy. *_Forest Drainage Engineering - A Design Manual_*. Ireland: COFORD, 1999.
- Ryan, T., H. Phillips, J. Ramsey and J. Dempsey. *_Forest Road Manual: Guidelines for the design, construction and management of forest roads_*. Dublin, Ireland: COFORD, 2004.

Requested Resources

- Lecture Room: Loose Seated